

Q1

Q1 What is the difference between while, do-while, and for loops? Explain with examples.

- While loop is entry control loop. it is execute when certain condition is true.

E.g 👍

```
#include <stdio.h>
```

```
Int main(){
```

```
    int i=1;
    while(i<=10){
        printf("%d\n",i);
        i++;
    }
```

```
    return 0;
}
```

- Do-while loop is exit control loop. it is excute at least onece if condition is true /false.

E.g 👍

```
#include <stdio.h>
```

```
Int main(){
```

```
    Int i= 1;
```

```
    Do{
        printf("%d\n",i);
        i++;
    }while(i<=10);
```

```
    return 0;
```

```
}
```

- For loop is a loop it is execute when certain condition is true. It is working on iterations.

E.g 👍

```
#include <stdio.h>
```

```
Int main()
```

```
{
```

```
Int i,n=10;
```

```
for(i=0;i<n;i++){
```

```
printf("%d\n",i);
```

```
}
```

```
return 0;
```

```
}
```

Q3

Q3. Write a C program to find the largest element in an array.

Input:

5

10 20 5 40 15

Output:

40

```
#include <stdio.h>
```

```
int main(){
    int arr[5] = {10,20,5,40,15};
    int max,i,n=5;

    max = arr[0];

    for(i=1;i<n;i++){
        if(arr[i] > max){
            max = arr[i];
        }
    }
    printf("max number is = %d",max);
    return 0;
}
```

Q4

Q4 . Explain pointers in C. What are the operators & and *? Give an example.

Pointer is a variable to hold another variable address in memory it is called pointer .

The & operator is denoted the address of the value to store .

The * operator is denoted to point this address value is store in variable.

E.g 👍

```
#include <stdio.h>
```

```
void swap(int *a, int *b){
```

```
    int temp;
```

```
    temp = *a;
```

```
    *a= *b;
```

```
    *b= temp;
```

```
}
```

```
int main(){
```

```
    int num1,num2;
```

```
    printf("Enter first number :");
```

```
    scanf("%d",&num1);
```

```
    printf("Enter first number :");
```

```
    scanf("%d",&num2);
```

```
    swap(&num1,&num2);
```

```
    printf("After Swapping :\n");
```

```
    printf("a = %d\n b= %d\n",num1,num2);
```

```
    return 0;
```

```
}
```

Q5

Q5 Write a C program to count the number of positive integers in an array.

Input:

6

-2 4 -1 8 0 3

```
#include <stdio.h>
```

```
int arr[6] = {-2, 4, -1, 8, 0, 3};
```

```
int count=0,i,n=6;
```

```
int main(){
```

```
    for(i=0;i<n;i++){
```

```
        count++;
```

```
    }
```

```
    printf("Total count is = %d",count);
```

```
    return 0;
```

```
}
```